



Sequences and Series · Mark Scheme

Chapter 4: Sequences and Series. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Arithmetic

- 1 $u_{15} = 7 + 14(4)$ M1 = 63 A1.
- 2 $a + 2d = 11, a + 7d = 31$ M1; $5d = 20 \Rightarrow d = 4$ A1; $a = 3$ A1.
- 3 $a = 4, d = 5$ B1; $S_{25} = \frac{25}{2}[8 + 24(5)]$ M1 = 1600 A1.
- 4 First term 54, last term 246 B1; $246 = 54 + 6(n-1) \Rightarrow n = 33$ M1 A1; $S = \frac{33}{2}(54 + 246) = 4950$ A1.
- 5(a) $100 + 11(15)$ M1 = £265 A1.
- 5(b) $n = 24$ B1; $S_{24} = \frac{24}{2}[200 + 23(15)]$ M1 = £6540 A1.
- 5(c) $\frac{n}{2}[200 + 15(n-1)] > 10000$ M1; $15n^2 + 185n - 20000 > 0 \Rightarrow 3n^2 + 37n - 4000 > 0$ (given) A1; critical value $n = \frac{-37 + \sqrt{37^2 + 48000}}{6} = 30.87 \dots$ M1; so **31 months** A1.
Check: $S_{30} = 9525, S_{31} = 10075$. Common error: rounding 30.87 down to 30.

Section B – Geometric

- 6 $u_6 = 5 \times 3^5$ M1 = 1215 A1.
- 7 (a) $S_8 = \frac{64(1 - (\frac{1}{2})^8)}{1 - \frac{1}{2}}$ M1 = 127.5 A1. (b) $S_{\infty} = \frac{64}{1 - \frac{1}{2}} = 128$ B1.
- 8 $ar = 12, ar^4 = -96$ M1; $r^3 = -8 \Rightarrow r = -2$ A1; $a = -6$ A1.
- 9 (a) $\frac{a}{1-r} = 4a$ M1; $r = \frac{3}{4}$ A1. (b) $a(\frac{3}{4})^2 = 18$ M1; $a = 32$ A1.
- 10(a) $ar = 48$ B1; $\frac{a}{1-r} = 256$ B1; substitute $a = \frac{48}{r}$: $48 = 256r(1-r)$ M1; $256r^2 - 256r + 48 = 0 \Rightarrow 16r^2 - 16r + 3 = 0$ (given) A1.
- 10(b) $(4r-1)(4r-3) = 0 \Rightarrow r = \frac{1}{4}, \frac{3}{4}$ M1; $a = 192$ or $a = 64$ A1.
- 10(c) $S_{\infty} - S_5 = \frac{ar^5}{1-r} = 256(\frac{3}{4})^5$ M1 = 60.75 A1.

Section C – Sigma notation

- 11 $3 + 5 + 7 + 9 + 11$ M1 = 35 A1.
- 12 Arithmetic, $a = 2, l = 97, n = 20$ M1 A1; $S = 10(2 + 97) = 990$ A1.
- 13 $\sum_{r=1}^{20} (3r+1)$ M1 A1 (general term, limits); $S = 10(4 + 61) = 650$ A1.
- 14 Geometric, $a = 6, \text{ratio } 2, n = 8$ M1; $S = \frac{6(2^8 - 1)}{2 - 1}$ M1 = 1530 A1.
Common error: taking the first term as 3 (i.e. starting from $r = 0$).
- 15(a) Arithmetic with $a = 5, d = 4$ B1; $\frac{n}{2}[10 + 4(n-1)]$ M1 = $\frac{n}{2}(4n+6) = n(2n+3)$ A1.
- 15(b) $S_{30} - S_{10} = 30(63) - 10(23)$ M1 = 1660 A1. Common error: subtracting S_{11} .
- 15(c) $2k^2 + 3k - 377 = 0 \Rightarrow (k-13)(2k+29) = 0$ M1; $k = 13$ A1.



Section D – Recurrence relations

- 16** $u_2 = 10, u_3 = 28, u_4 = 82$ M1 A1.
- 17** (a) 5.5, 5.75, 5.875 M1 A1. (b) Increasing, converging towards 6 B1.
- 18** (a) $u_2 = \frac{2}{3}, u_3 = -\frac{1}{2}, u_4 = 3$ M1; periodic with period 3 A1. (b) $100 = 3 \times 33 + 1$, so $u_{100} = u_1 = 3$ B1.
- 19** $u_2 = 2k + 3$ B1; $u_3 = k(2k + 3) + 3 = 17$ M1; $2k^2 + 3k - 14 = 0 \Rightarrow (2k + 7)(k - 2) = 0$ M1; $k = 2$ or $k = -\frac{7}{2}$ A1.
- 20(a)** $a_2 = 10 - c$ B1; $a_3 = 2(10 - c) - c = 20 - 3c$ B1.
- 20(b)** $5 + (10 - c) + (20 - 3c) = 11$ M1 A1 ft; $35 - 4c = 11 \Rightarrow c = 6$ A1.
- 20(c)** $a_4 = -2, a_5 = -10$ M1 A1; decreasing (each term smaller than the last).

Section E – Binomial (positive integer powers)

- 21** $1 + 4x + 6x^2 + 4x^3 + x^4$ M1 A1.
- 22** $2^6 + \binom{6}{1}2^5(-x) + \binom{6}{2}2^4(-x)^2$ M1; $64 - 192x + 240x^2$ A1 A1.
- 23** $\binom{7}{3}3^4(2x)^3$ M1 A1; coefficient = $35 \times 81 \times 8 = 22680$ A1.
- 24** $\binom{8}{2}k^2 = 7 \times \binom{8}{1}k$ M1; $28k^2 = 56k \Rightarrow k = 2$ A1; $\binom{8}{3}2^3$ M1 = 448 A1.
- 25(a)** $1 + 10\left(-\frac{x}{2}\right) + \binom{10}{2}\left(-\frac{x}{2}\right)^2 + \binom{10}{3}\left(-\frac{x}{2}\right)^3$ M1; $1 - 5x + \frac{45}{4}x^2 - 15x^3$ A1 A1.
- 25(b)** $1 - \frac{x}{2} = 0.995 \Rightarrow x = 0.01$ M1; 0.95111 A1.
- 25(c)** $3 \times \frac{45}{4} + 2 \times (-5)$ M1 A1 ft; $= \frac{95}{4}$ A1.

Section F – Binomial (negative and fractional powers)

- 26** $1 - x + x^2 - x^3$ M1 A1; valid for $|x| < 1$ B1.
- 27** $1 + \frac{1}{2}(3x) + \frac{\frac{1}{2}(-\frac{1}{2})}{2!}(3x)^2 + \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{3!}(3x)^3$ M1; $1 + \frac{3}{2}x - \frac{9}{8}x^2 + \frac{27}{16}x^3$ A1 A1; valid for $|x| < \frac{1}{3}$ B1.
- 28** $(4 - x)^{-\frac{1}{2}} = 4^{-\frac{1}{2}}\left(1 - \frac{x}{4}\right)^{-\frac{1}{2}}$ M1 B1 (for $\frac{1}{2}$); $\left(1 - \frac{x}{4}\right)^{-\frac{1}{2}} = 1 + \frac{x}{8} + \frac{3}{128}x^2$ M1 A1; $\frac{1}{\sqrt{4-x}} \approx \frac{1}{2} + \frac{1}{16}x + \frac{3}{256}x^2$ A1.
Common error: writing $4^{-\frac{1}{2}} = 2$ or forgetting to multiply back through.
- 29** $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3$ M1 A1; $(1 + 2x)(1 + 2x + 3x^2 + 4x^3)$ M1; $= 1 + 4x + 7x^2 + 10x^3$ A1.
- 30(a)** $1 + \frac{1}{2}(-8x) + \frac{\frac{1}{2}(-\frac{1}{2})}{2}(-8x)^2 + \frac{\frac{1}{2}(-\frac{1}{2})(-\frac{3}{2})}{6}(-8x)^3$ M1 M1; $1 - 4x - 8x^2 - 32x^3$ A1 A1.
- 30(b)** $|x| < \frac{1}{8}$ B1.
- 30(c)** $\sqrt{1 - \frac{8}{100}} = \sqrt{\frac{92}{100}} = \frac{2\sqrt{23}}{10} = \frac{\sqrt{23}}{5}$ B1; $\sqrt{23} \approx 5(1 - 0.04 - 0.0008 - 0.000032)$ M1 = 4.79584 A1.
(True value 4.79583..., which shows how quickly the series converges for small x .)

Roar-level challenge

- 31** $(1 + 2x)^5 = 1 + 10x + 40x^2 + \dots$ M1 A1; $(1 - x)^{-1} = 1 + x + x^2 + \dots$ B1; coefficient of x^2 : $40 + 10 + 1 = 51$ A1.